Day35_LoggingAlerting

Q1 — Fundamentals [2 Marks]

Explain the difference between Logging, Monitoring and Alerting.

- **Logging**: Recording application activities and events for future investigation and reference.
- **Monitoring**: Continuously observing logs and system activities to identify unusual behaviour or security issues.
- **Alerting**: Sending notifications to security teams when suspicious activities or predefined security conditions are detected.

Q2 — Security Events [3 Marks]

List at least six security-relevant application events that should be considered for logging. Explain why any two are important.

1. Failed login attempts.
2. Password reset requests.
3. Account lockouts.
4. Privilege or role changes.
5. Access denied events.
6. MFA failures.
7. Logout events.

Why two are important:

- **Failed login attempts:** Help identify brute-force attacks and suspicious login activities.
- **Privilege changes:** Help detect unauthorized attempts to gain administrative access.

Q3 — Sensitive Data [2 Marks]

List at least five examples of sensitive information that should normally not appear in application logs.

1. Plaintext passwords.
2. Authentication tokens.
3. Session cookies.
4. MFA codes and secrets.
5. API keys.

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Q4 — Log Integrity [2 Marks]

Why should security logs be protected from unauthorized modification or deletion? Give two controls.

Security logs must be protected from unauthorized modification or deletion because attackers could remove evidence, hide suspicious activities or alter investigation records.

Two security controls:

1. **Access Control:** Restrict log access and modification permissions to authorized personnel.
2. **Log Integrity Protection:** Use cryptographic hashes or digital signatures to detect unauthorized modifications.

Q5 — Alerting [3 Marks]

Design one simple alert rule for detecting suspicious repeated authentication failures. Include Condition, Time Window, Severity and Recommended Action.

- **Condition**: More than 5 failed login attempts from the same source IP.
- **Time** Window: Within 5 minutes.
- **Severity**: HIGH.
- **Recommended Action:** Generate an alert, notify the security team, investigate the source IP and monitor targeted accounts.

Q6 — False Positives [2 Marks]

Explain False Positive and Alert Fatigue. Why can excessive alerts reduce security effectiveness?

- **False Positive**: An alert generated when normal or legitimate activity is incorrectly identified as suspicious.
- **Alert Fatigue:** Occurs when security teams receive too many alerts and become overwhelmed, making it difficult to identify genuine threats.

Q7 — Practical Lab [4 Marks]

Using the supplied Day 35 training log: identify the suspicious source, failure count, time window, and provide a short investigation summary. Attach screenshots.

Evidences:

Day35_LoggingAlerting

```
┌──(kirthikaa㉿kali)-[~/Desktop]
└─$ cat day35_auth.log
2026-10-09T10:00:01+05:30 | LOGIN_SUCCESS | user=student01 | source=192.0.2.10 | result=success
2026-10-09T10:00:45+05:30 | LOGIN_SUCCESS | user=student02 | source=192.0.2.14 | result=success
2026-10-09T10:02:12+05:30 | LOGIN_SUCCESS | user=student03 | source=192.0.2.18 | result=success
2026-10-09T10:03:07+05:30 | LOGOUT | user=student02 | source=192.0.2.14 | result=success
2026-10-09T10:04:33+05:30 | LOGIN_FAILED | user=student04 | source=198.51.100.77 | result=invalid_credentials
2026-10-09T10:05:10+05:30 | LOGIN_SUCCESS | user=student05 | source=192.0.2.22 | result=success
2026-10-09T10:06:41+05:30 | LOGIN_SUCCESS | user=student01 | source=192.0.2.10 | result=success
2026-10-09T10:08:02+05:30 | PASSWORD_RESET_REQUEST | user=student04 | source=192.0.2.31 | result=email_sent
2026-10-09T10:09:18+05:30 | LOGIN_SUCCESS | user=student06 | source=192.0.2.40 | result=success
2026-10-09T10:10:00+05:30 | LOGIN_FAILED | user=student05 | source=198.51.100.77 | result=invalid_credentials
2026-10-09T10:11:24+05:30 | LOGIN_SUCCESS | user=student07 | source=192.0.2.44 | result=success
2026-10-09T10:12:55+05:30 | LOGOUT | user=student03 | source=192.0.2.18 | result=success
2026-10-09T10:14:02+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:14:20+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:14:41+05:30 | LOGIN_FAILED | user=student02 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:03+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:29+05:30 | LOGIN_FAILED | user=student03 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:51+05:30 | LOGIN_FAILED | user=student02 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:16:12+05:30 | LOGIN_FAILED | user=admin_trainee | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:16:40+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:05+05:30 | LOGIN_FAILED | user=admin_trainee | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:31+05:30 | LOGIN_FAILED | user=student04 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:58+05:30 | LOGIN_FAILED | user=student05 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:18:22+05:30 | LOGIN_FAILED | user=student03 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:18:55+05:30 | LOGIN_FAILED | user=student06 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:20:14+05:30 | ACCESS_DENIED | user=student04 | source=192.0.2.31 | result=insufficient_permissions
```

```
┌──(kirthikaa㉿kali)-[~/Desktop]
└─$ grep "LOGIN_FAILED" day35_auth.log | awk -F'|' '{print $3}' | sort | uniq -c | sort -nr
      4  user=student01
      2  user=student06
      2  user=student05
      2  user=student04
      2  user=student03
      2  user=student02
      2  user=admin_trainee
```

```
┌──(kirthikaa㉿kali)-[~/Desktop]
└─$ grep "LOGIN_FAILED" day35_auth.log | grep "203.0.113.25"
2026-10-09T10:14:02+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:14:20+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:14:41+05:30 | LOGIN_FAILED | user=student02 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:03+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:29+05:30 | LOGIN_FAILED | user=student03 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:51+05:30 | LOGIN_FAILED | user=student02 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:16:12+05:30 | LOGIN_FAILED | user=admin_trainee | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:16:40+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:05+05:30 | LOGIN_FAILED | user=admin_trainee | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:31+05:30 | LOGIN_FAILED | user=student04 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:58+05:30 | LOGIN_FAILED | user=student05 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:18:22+05:30 | LOGIN_FAILED | user=student03 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:18:55+05:30 | LOGIN_FAILED | user=student06 | source=203.0.113.25 | result=invalid_credentials

┌──(kirthikaa㉿kali)-[~/Desktop]
└─$ grep "LOGIN_FAILED" day35_auth.log | grep "203.0.113.25" | head -1
2026-10-09T10:14:02+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials

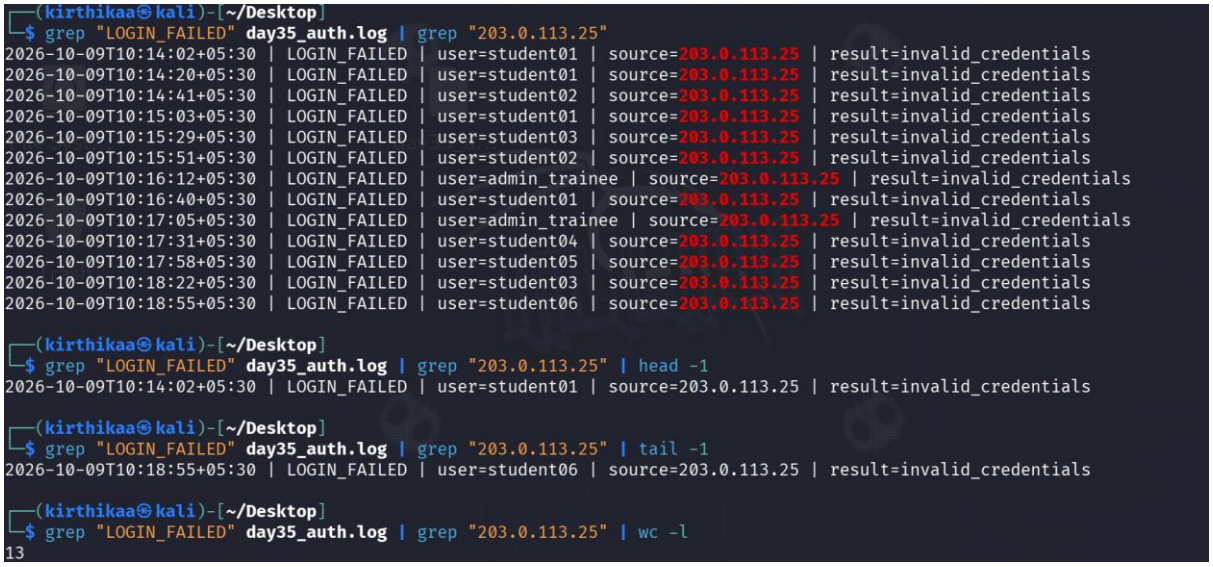
┌──(kirthikaa㉿kali)-[~/Desktop]
└─$ grep "LOGIN_FAILED" day35_auth.log | grep "203.0.113.25" | tail -1
2026-10-09T10:18:55+05:30 | LOGIN_FAILED | user=student06 | source=203.0.113.25 | result=invalid_credentials

┌──(kirthikaa㉿kali)-[~/Desktop]
└─$ grep "LOGIN_FAILED" day35_auth.log | grep "203.0.113.25" | wc -l
13
```

```
┌──(kirthikaa㉿kali)-[~/Desktop]
└─$ grep "LOGIN_FAILED" day35_auth.log | wc -l
16
```

```
┌──(kirthikaa㉿kali)-[~/Desktop]
└─$ grep "203.0.113.25" day35_auth.log
2026-10-09T10:14:02+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:14:20+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:14:41+05:30 | LOGIN_FAILED | user=student02 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:03+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:29+05:30 | LOGIN_FAILED | user=student03 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:51+05:30 | LOGIN_FAILED | user=student02 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:16:12+05:30 | LOGIN_FAILED | user=admin_trainee | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:16:40+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:05+05:30 | LOGIN_FAILED | user=admin_trainee | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:31+05:30 | LOGIN_FAILED | user=student04 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:58+05:30 | LOGIN_FAILED | user=student05 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:18:22+05:30 | LOGIN_FAILED | user=student03 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:18:55+05:30 | LOGIN_FAILED | user=student06 | source=203.0.113.25 | result=invalid_credentials
```

Day35_LoggingAlerting



Q8 — Secure Logging Design [2 Marks]

Provide at least five recommendations for a secure application logging and alerting system.

- Record important security events, including login attempts and access-control changes.
- Avoid logging sensitive information such as passwords, tokens and API keys.
- Protect log files using access controls and encryption.
- Implement real-time monitoring and alerts for suspicious activities.
- Regularly review logs and maintain backups to support investigations and recovery.